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Project Report

ON

ANOMALY DETECTION FOR HEALTHCARE TRANSACTIONS

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EXECUTIVE SUMMARY

The Problem in Healthcare

Healthcare fraud costs the medical industry billions annually through fake services and overcharging. Insurance companies currently rely on basic, hardcoded computer rules (like flagging claims over $1,000) to stop it. However, fraudsters easily bypass these rules by simply billing $999 to avoid detection.

The AI-Powered Solution

This project introduces the Healthcare Fraud Anomaly Detection System, an automated AI platform that learns "normal" billing behavior instead of relying on static rules. When a claim arrives, the system performs two simultaneous checks:

- **Document Check:** An Isolation Forest model evaluates the claim's static details (patient age, diagnosis, cost) for logical consistency.
- **Timeline Check:** An LSTM Autoencoder analyzes the doctor's past billing history to detect sudden, suspicious changes in their normal routine.

How the System Works for Users

By mathematically fusing these checks, the system categorizes incoming claims into three actionable zones: **Normal** (safe to pay), **Suspicious** (human review required), and **High Risk** (stop payment). Auditors manage this workflow through a clean Streamlit web dashboard supporting both instantaneous single-claim analysis and massive CSV bulk uploads.

Key Project Achievements

During the final testing phase, the system successfully met its major technical and business goals:

- **High Accuracy & Precision:** Achieved 93.00% overall accuracy on heavily unbalanced data and an elite 98.44% precision score, virtually eliminating wasted time on false alarms. **Explainable**
- **AI:** Replaces the AI "black box" with SHAP (SHapley Additive exPlanations). It generates simple visual bar charts showing exactly which details triggered the alarm, allowing human auditors to legally and confidently verify the computer's decisions.

1. BACKGROUND

Fraud detection systems in healthcare commonly utilize rule-based approaches, supervised classification, or anomaly detection. Rule-based systems are easy to interpret but are entirely rigid, failing to generalize well when fraudulent behavior patterns adapt. Supervised classification depends heavily on perfectly labeled fraud datasets, which are largely unavailable, proprietary, or incomplete in real-world scenarios. Anomaly detection serves as the most practical approach when real fraud labels are limited, as it mathematically learns "normal" baseline behavior and identifies severe deviations from that learned standard.

In a conventional healthcare setup, suspicious claims are manually reviewed using basic heuristics. This is intensely time-consuming, scales poorly during massive batch audits, and ignores provider-level temporal behavior changes by evaluating claims in isolation. This project bridges this gap by merging historical provider context with machine learning, transitioning from static manual review to automated, algorithmic screening.

1.1 Aim

The primary aim of this project is to build a hybrid healthcare anomaly detection system that identifies suspicious claims automatically to reduce unnecessary manual auditing effort. By combining static and temporal fraud indicators into a single pipeline, the system evaluates both individual claim features and sequential provider billing behaviors simultaneously.

Furthermore, the project aims to provide an intuitive Graphical User Interface (GUI) to support both real-time single-claim analysis and high-throughput batch CSV audit reviews, while using Explainable AI (SHAP) to eliminate the "black box" nature of algorithmic decisions.

1.2 Technologies

The system was engineered utilizing a modern Python-based microservice tech stack:

- **Machine Learning:** Scikit-Learn(Isolation Forest) and TensorFlow/Keras(LSTM Autoencoder).
- **Explainable AI (XAI):** SHAP (SHapley Additive exPlanations).
- **Backend Framework:** FastAPI for RESTful API routing and data validation.
- **Frontend Framework:** Streamlit for interactive dashboard deployment.
- **Database & Processing:** SQLite for historical persistence, and Pandas/NumPy for feature engineering.

1.3 Hardware Architecture

The system is designed to operate efficiently without requiring specialized enterprise-grade servers or expensive dedicated GPUs. The architecture utilizes standard x86-64 multi-core CPU infrastructure to handle parallel batch inference requests. A minimum of 8GB of System RAM is required to maintain the in-memory state of the loaded neural network model weights and process large Pandas dataframes during batch CSV uploads.

1.4 Software Architecture

The software relies on a decoupled, four-tier microservice paradigm:

- **Presentation Tier:** Hosted in Streamlit, managing user inputs and visually rendering SHAP charts and risk zones.
- **Application Tier:** Hosted in FastAPI, acting as the central router to validate incoming JSON requests and manage HTTP responses.
- **Machine Learning Tier:** Contains the feature engineering scripts and the pre-trained weights for the Isolation Forest and LSTM models.
- **Data Tier:** Contains the persistent SQLite relational database for retrieving historical provider billing context.

2. SYSTEM

This chapter details the specific requirements, logical architecture, implementation phases, and evaluation metrics used to construct and validate the anomaly detection platform.

2.1 Requirements

2.1.1 Functional requirements

The system must accept a single healthcare claim payload via API for live evaluation and support batch CSV uploads for bulk review. It must autonomously load provider-specific historical context from the database, compute hybrid anomaly scores, and classify claims into operational risk zones (Normal, Suspicious, High Risk). Finally, it must generate natural language explanation text for anomalous outputs and allow users to export finalized batch reports.

2.1.2 User requirements

Human auditors require an interface that is visually accessible without deep technical expertise. The system must abstract complex mathematical tensors into simple UI elements, providing clear color-coded alerts (Green, Yellow, Red) and distinct action-oriented summaries (e.g., "Auto-clear", "Review", "Escalate").

2.1.3 Environmental requirements

The platform must be reproducible across diverse execution environments. It requires a standard Python 3.9+ runtime and assumes containerized deployment using Docker to guarantee that internal microservice networking, database routing, and library dependencies remain completely isolated from the host operating system.

2.2 Design and Architecture

The architectural design utilizes a dual-pronged approach to anomaly detection. When raw JSON data enters the pipeline, the Feature Engine dynamically constructs two distinct structures: a spatial 1D array for static features (e.g., billing deviation, service duration) and a sequential 3D tensor window spanning the last seven transactions for temporal features (e.g., billing velocity, time gaps).

These structures are passed to the Hybrid Inference Engine. The Isolation Forest evaluates the static array, while the LSTM Autoencoder calculates the Mean Squared Error (MSE) reconstruction loss of the temporal sequence. Both outputs are z-score normalized and mathematically fused into a final unified Anomaly Score before a strict percentile threshold is applied to dictate the final system decision.

2.3 Implementation

Implementation systematically constructed the decoupled tiers. The Data Layer utilized a custom SQLite loader to handle full dataset extraction during model training and optimized lookups during live inference. Feature engineering logic dynamically scaled features using a `RobustScaler` to prevent data leakage and handle extreme financial outliers natively.

The ML models were trained offline, and their artifacts (`.pkl` and `.keras`) were frozen and exported. The API implementation was achieved via FastAPI, enforcing strict variable types for incoming JSON payloads using Pydantic. Finally, the Streamlit frontend was integrated via REST HTTP requests, utilizing session-state management to preserve expensive batch operations without crashing the browser.

2.4 Testing

Testing was executed across the entire software stack. Backend integration tests verified that the FastAPI server could correctly fetch provider histories from SQLite without throwing 500 Internal Server Errors. The ML integration was verified by bombarding the endpoints with data to ensure the fusion engine consistently outputted correctly bounded Anomaly Scores. Performance tests verified that the multi-core API could successfully ingest and process 150-row CSV batch audits without timing out.

2.5 Graphical User Interface (GUI) Layout

GUI testing verified that the Streamlit dashboard effectively routed users between two distinct operational workflows. The Single Claim Analysis layout was tested to ensure the input form dynamically responded to data entry and cleanly rendered SHAP feature contribution charts without breaking page bounds. The Batch Audit layout was evaluated to ensure large CSV files parsed correctly into interactive, paginated data tables, providing visual flags for escalated claims.

2.6 Customer testing

Simulated customer acceptance testing focused on the clarity of the AI explanations. The system was validated to ensure it generated context-aware reasons (e.g., "Claim amount is materially above the provider baseline") rather than outputting confusing raw mathematics. Error boundaries were also tested; deliberately uploading broken CSV files correctly triggered a polite UI warning rather than crashing the application backend.

2.7 Evaluation

The system was rigorously tested to verify algorithmic stability and resilience against unbalanced datasets.

Threshold Tuning & Stability: The hybrid scoring threshold was empirically tuned to the 92nd percentile, which improved classification performance. Across five automated test runs utilizing different random seeds (Seeds 123, 132, 451, 143, 389), the system yielded a highly stable average binary accuracy of 90.93%.

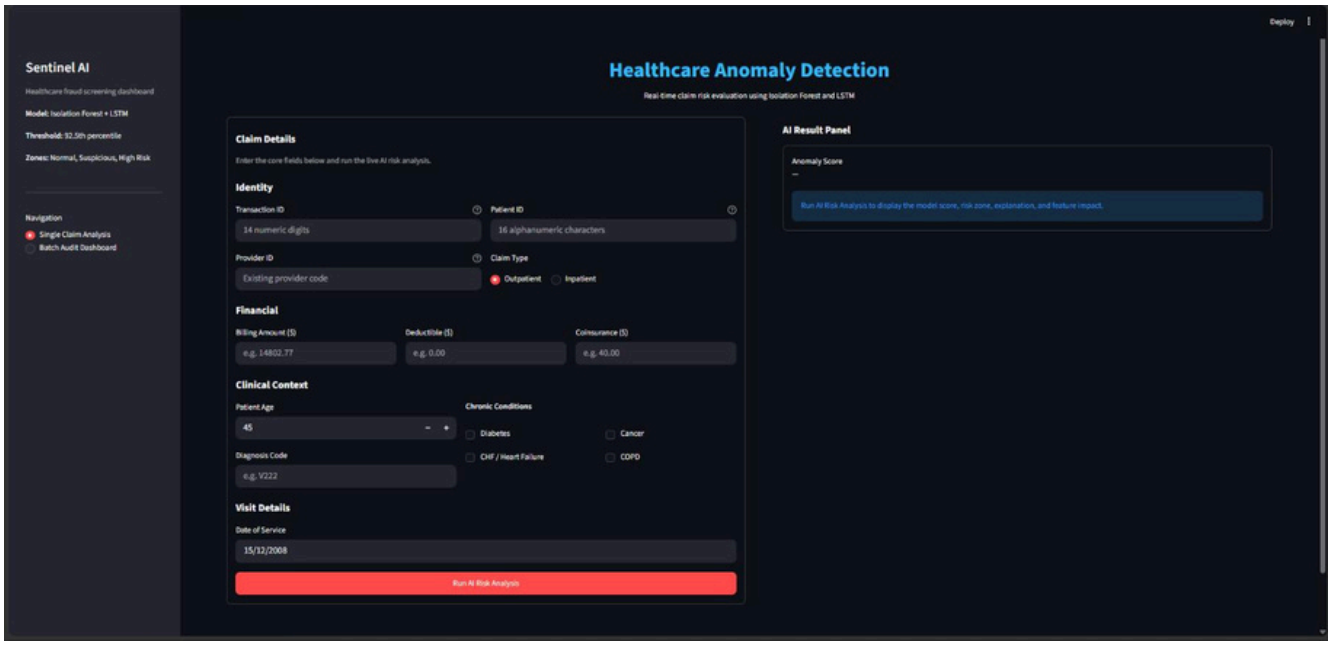
Imbalanced Stress Testing: Because fraud datasets are rarely 50/50 in the real world, the model was tested using severely uneven class splits (e.g., 30 normal claims to 70 anomalies, and vice versa). Across three distinct imbalanced tests, the model's average accuracy rose to 93.00%. More importantly, the system achieved a peak Precision of 98.44%. The incredibly high precision dictates that when the system flags a claim as an anomaly, it is almost certainly correct, minimizing false alarms and reviewer fatigue.

Batch Metrics: In a standardized 150-row batch test, the system achieved 92.00% Accuracy, 97.01% Precision, 86.67% Recall, and a False Positive Rate of only 2.67%, proving its operational readiness for high-volume auditing.

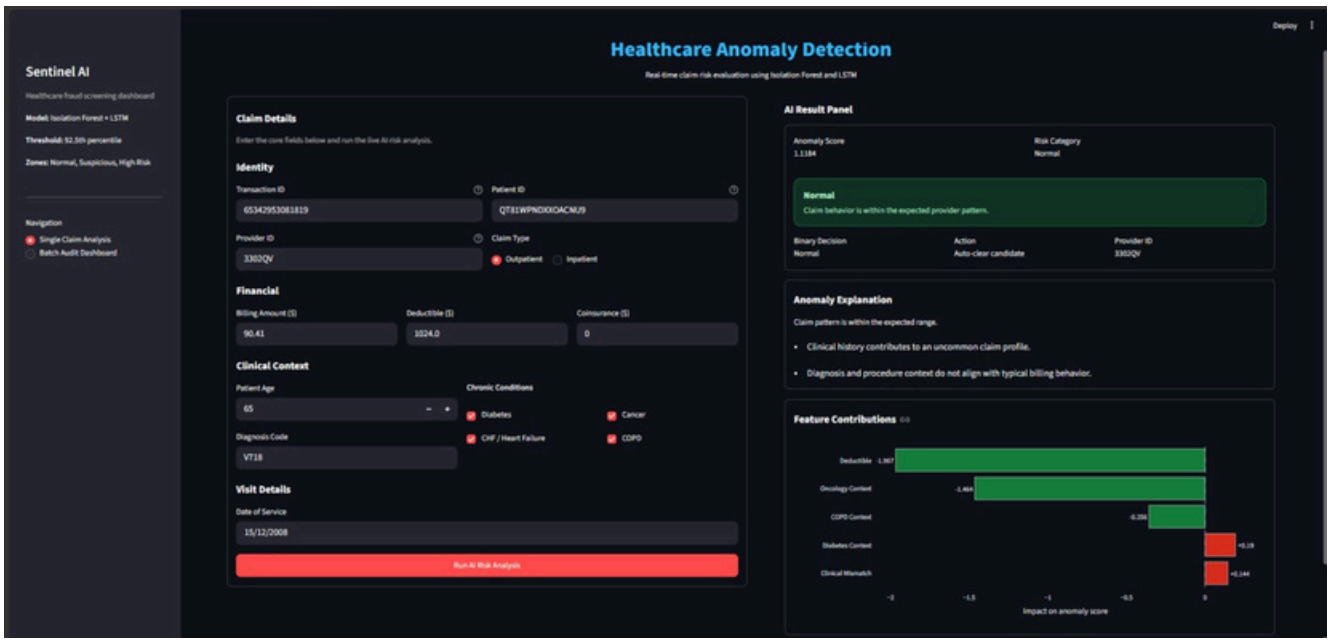
3. SNAPSHOTS OF THE PROJECT

The following technical captures document the Graphical User Interface layout, data presentation states, and the live terminal execution of the microservice backend.

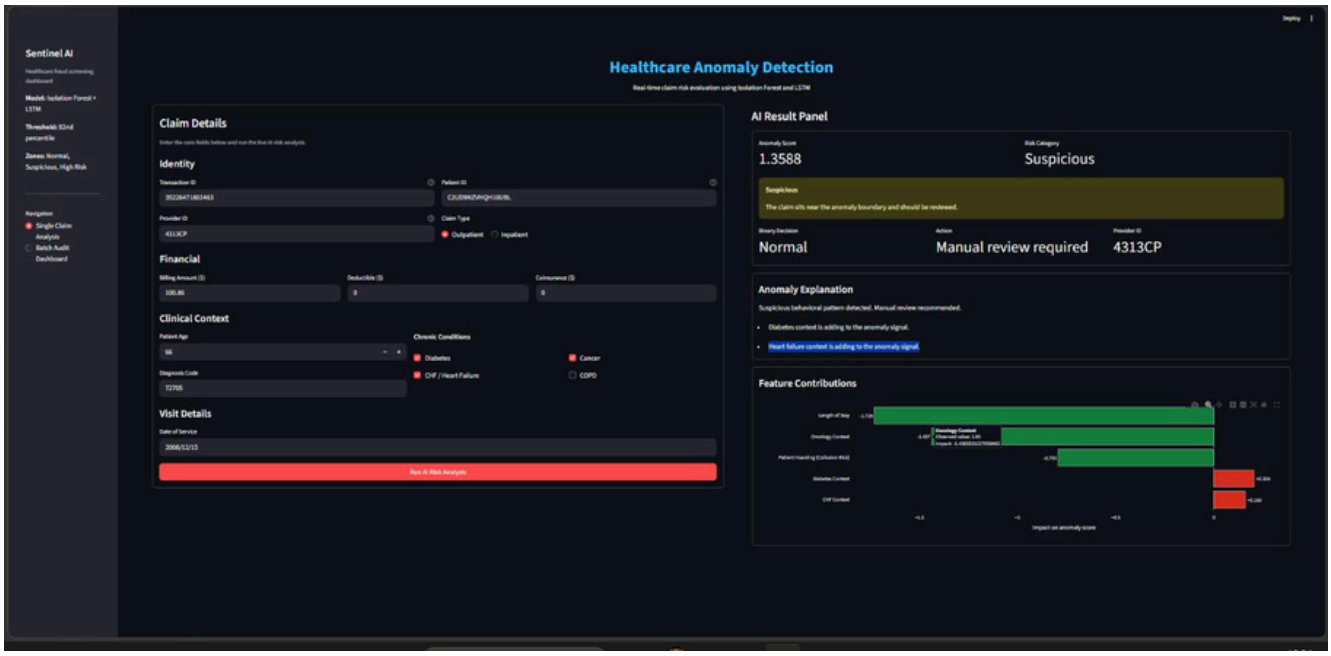
Initial Input Interface (Empty Form)



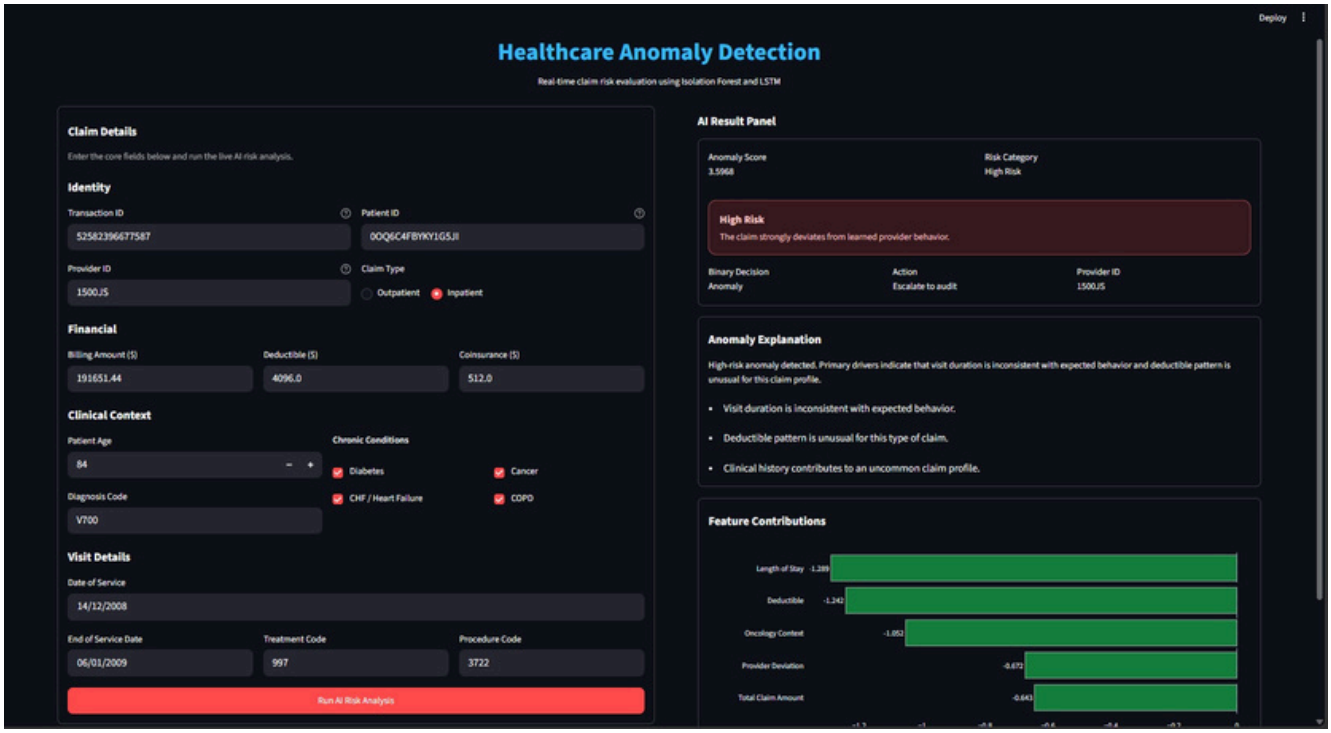
Normal Claim Behavior (Auto-Clear Candidate)



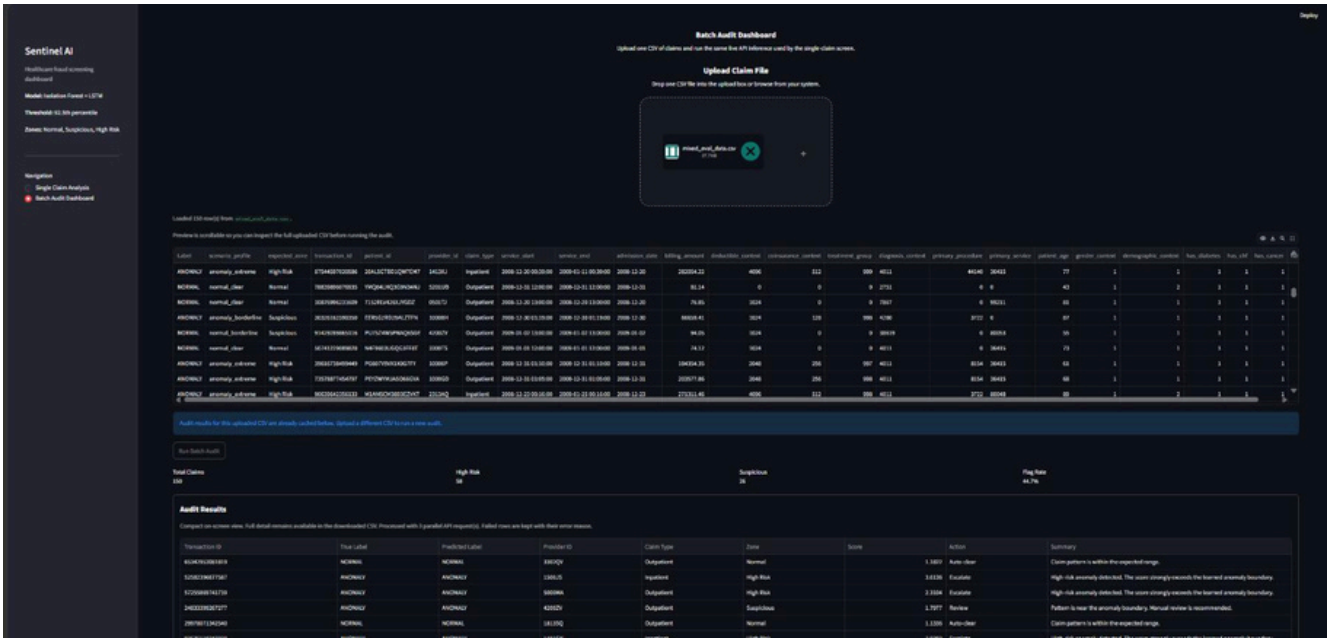
Suspicious Claim Behavior (Manual Review Recommended)



High Risk Anomaly Alert (Escalate to Audit)

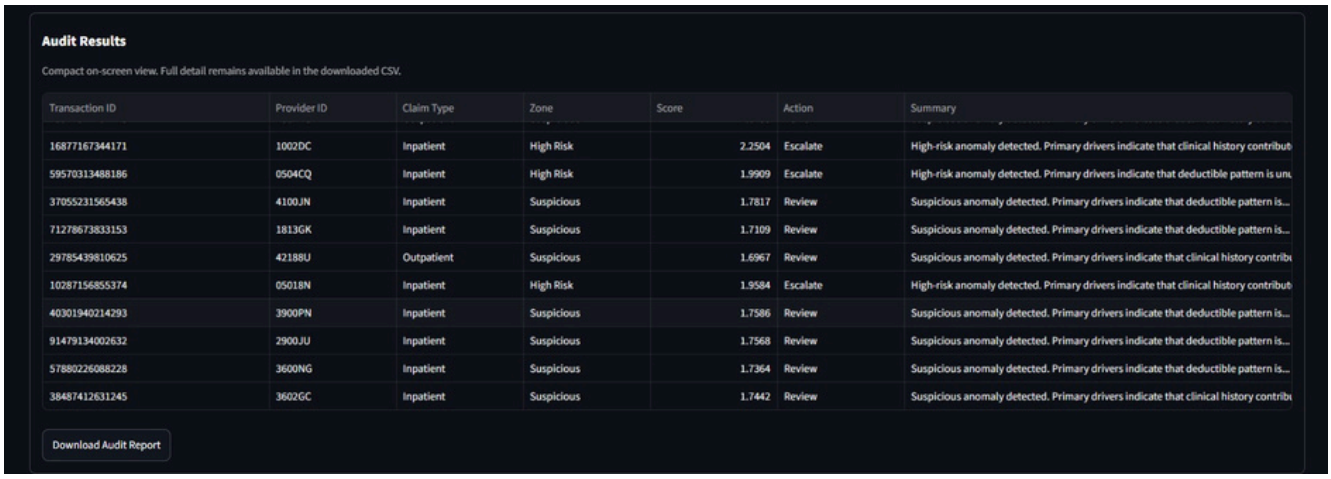


Batch Audit Dashboard Overview



Detailed Audit Results Table (Escalation & Review Flags)

Screenshot 2026-03-25 195701.png



Audit Results

Compact on-screen view. Full detail remains available in the downloaded CSV.

Transaction ID	Provider ID	Claim Type	Zone	Score	Action	Summary
16877167344171	1002DC	Inpatient	High Risk	2.2504	Escalate	High-risk anomaly detected. Primary drivers indicate that clinical history contribut
59570313488186	0504CQ	Inpatient	High Risk	1.9909	Escalate	High-risk anomaly detected. Primary drivers indicate that deductible pattern is unu
37055231565438	4100JN	Inpatient	Suspicious	1.7817	Review	Suspicious anomaly detected. Primary drivers indicate that deductible pattern is...
71278673833153	1813GK	Inpatient	Suspicious	1.7109	Review	Suspicious anomaly detected. Primary drivers indicate that deductible pattern is...
29785439810625	42188U	Outpatient	Suspicious	1.6967	Review	Suspicious anomaly detected. Primary drivers indicate that clinical history contribu
10287156855374	05018N	Inpatient	High Risk	1.9584	Escalate	High-risk anomaly detected. Primary drivers indicate that clinical history contribut
40301940214293	3900PN	Inpatient	Suspicious	1.7586	Review	Suspicious anomaly detected. Primary drivers indicate that deductible pattern is...
91479134002632	2900JU	Inpatient	Suspicious	1.7568	Review	Suspicious anomaly detected. Primary drivers indicate that deductible pattern is...
57880226088228	3600NG	Inpatient	Suspicious	1.7364	Review	Suspicious anomaly detected. Primary drivers indicate that deductible pattern is...
38487412631245	3602GC	Inpatient	Suspicious	1.7442	Review	Suspicious anomaly detected. Primary drivers indicate that clinical history contribu

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Terminal Log: Live Microservice Traffic & API Routing

```
backend-1   | Loading frozen models and baseline statistics for inference...
backend-1   | --> Received Outpatient Claim: Provider 3100GJ | Amount: $134.563230799348
backend-1   | --> Received Outpatient Claim: Provider 3100GJ | Amount: $140.2240288004539
backend-1   | --> Received Outpatient Claim: Provider 3100GJ | Amount: $124.09153460744103
backend-1   | INFO:     172.18.0.3:41900 - "POST /evaluate_transaction?include_explainability=false HTTP/1.1" 200 OK
backend-1   | INFO:     172.18.0.3:41888 - "POST /evaluate_transaction?include_explainability=false HTTP/1.1" 200 OK
backend-1   | INFO:     172.18.0.3:41910 - "POST /evaluate_transaction?include_explainability=false HTTP/1.1" 200 OK
backend-1   | --> Received Outpatient Claim: Provider 3100GJ | Amount: $125.60773236562238
backend-1   | --> Received Outpatient Claim: Provider 3100GJ | Amount: $124.74086723657749
backend-1   | --> Received Outpatient Claim: Provider 3100GJ | Amount: $141.4894688250035
backend-1   | INFO:     172.18.0.3:57850 - "POST /evaluate_transaction?include_explainability=false HTTP/1.1" 200 OK
backend-1   | --> Received Outpatient Claim: Provider 3100GJ | Amount: $148.16818828739662
backend-1   | INFO:     172.18.0.3:57864 - "POST /evaluate_transaction?include_explainability=false HTTP/1.1" 200 OK
backend-1   | --> Received Outpatient Claim: Provider 3100GJ | Amount: $126.00927608001184
backend-1   | INFO:     172.18.0.3:57852 - "POST /evaluate_transaction?include_explainability=false HTTP/1.1" 200 OK
backend-1   | --> Received Outpatient Claim: Provider 3100GJ | Amount: $134.25848249228255
backend-1   | INFO:     172.18.0.3:36484 - "POST /evaluate_transaction?include_explainability=false HTTP/1.1" 200 OK
backend-1   | --> Received Outpatient Claim: Provider 3100GJ | Amount: $145.9758214543003
backend-1   | INFO:     172.18.0.3:36466 - "POST /evaluate_transaction?include_explainability=false HTTP/1.1" 200 OK
backend-1   | --> Received Outpatient Claim: Provider 3100GJ | Amount: $1168.75
backend-1   | INFO:     172.18.0.3:36476 - "POST /evaluate_transaction?include_explainability=false HTTP/1.1" 200 OK
backend-1   | --> Received Inpatient Claim: Provider 3100GJ | Amount: $687.5
backend-1   | INFO:     172.18.0.3:58312 - "POST /evaluate_transaction?include_explainability=false HTTP/1.1" 200 OK
backend-1   | --> Received Outpatient Claim: Provider 3100GJ | Amount: $137.5
backend-1   | INFO:     172.18.0.3:58322 - "POST /evaluate_transaction?include_explainability=false HTTP/1.1" 200 OK
backend-1   | --> Received Outpatient Claim: Provider 3100GJ | Amount: $412.5
backend-1   | INFO:     172.18.0.3:58328 - "POST /evaluate_transaction?include_explainability=false HTTP/1.1" 200 OK
backend-1   | INFO:     172.18.0.3:50808 - "POST /evaluate_transaction?include_explainability=false HTTP/1.1" 200 OK
backend-1   | INFO:     172.18.0.3:50824 - "POST /evaluate_transaction?include_explainability=false HTTP/1.1" 200 OK
```

Terminal Log: Automated Batch Inference Metrics

```
Running inference on 150 rows from data/samples/mixed_eval_data.csv...

Batch inference complete in 239.16 seconds.
Binary Accuracy: 94.00%
Zone Accuracy: 50.67%

Actual vs Predicted:
 transaction_id provider_id   scenario_profile actual_label predicted_label expected_zone alert_zone  is_correct  zone_is_correct  anomaly_score
 65342953081819      3302QV  normal_borderline       NORMAL         NORMAL    Suspicious     Normal        True            False         1.1822
 52582396677587      1500JS    anomaly_extreme      ANOMALY        ANOMALY     High Risk  High Risk        True             True         3.6136
 57255869741739      5000MA    anomaly_extreme      ANOMALY        ANOMALY     High Risk  High Risk        True             True         2.3104
 24833399267277      4200ZV anomaly_borderline      ANOMALY        ANOMALY    Suspicious Suspicious        True             True         1.7977
 29978071342540      1813SQ  normal_borderline       NORMAL         NORMAL    Suspicious     Normal        True            False         1.1336
 93676118243230      1401GN    anomaly_extreme      ANOMALY        ANOMALY     High Risk  High Risk        True             True         3.0382
 85762480999333      4500DP anomaly_borderline      ANOMALY        ANOMALY    Suspicious Suspicious        True             True         1.8176
 27326650576774      0501TJ  normal_borderline       NORMAL         NORMAL    Suspicious     Normal        True            False         1.0410
 47157434122090      10018S       normal_clear       NORMAL         NORMAL        Normal     Normal        True             True         0.1259
 27707615474401      1100DS anomaly_borderline      ANOMALY         NORMAL    Suspicious Suspicious       False             True         1.6155
 93444267955749      2300ZD    anomaly_extreme      ANOMALY        ANOMALY     High Risk  High Risk        True             True         1.9088
 86852126308236      4500DP       normal_clear       NORMAL         NORMAL        Normal Suspicious        True            False         1.5580
 82540548887196      4500DP  normal_borderline       NORMAL         NORMAL    Suspicious Suspicious        True             True         1.4373
 83192891859900      3300TS    anomaly_extreme      ANOMALY        ANOMALY     High Risk Suspicious        True            False         1.8094
 23584999174760      1401ZT    anomaly_extreme      ANOMALY         NORMAL     High Risk Suspicious       False            False         1.6324
 68649281504255      1413XJ anomaly_borderline      ANOMALY        ANOMALY    Suspicious  High Risk        True            False         2.9273
 54898514406928      1100DS       normal_clear       NORMAL         NORMAL        Normal     Normal        True             True         0.0606
 12981812188007      0504VD       normal_clear       NORMAL         NORMAL        Normal     Normal        True             True         1.0141
 94662983314272      3600YM  normal_borderline       NORMAL         NORMAL    Suspicious     Normal        True            False         0.0909
 14805785121579      1100XA  normal_borderline       NORMAL         NORMAL    Suspicious     Normal        True            False         1.1512
 21783446239572      10008H       normal_clear       NORMAL         NORMAL        Normal     Normal        True             True         1.1486
 48535117093674      3400PS    anomaly_extreme      ANOMALY        ANOMALY     High Risk Suspicious        True            False         1.8044
 23474364382392      1400JN       normal_clear       NORMAL         NORMAL        Normal     Normal        True             True         0.2871
 47486206031162      3302QV    anomaly_extreme      ANOMALY        ANOMALY     High Risk  High Risk        True             True         3.0151
 39560739745618      01008A  normal_borderline       NORMAL         NORMAL    Suspicious     Normal        True            False        -0.0733
 45868224237143      01008A       normal_clear       NORMAL         NORMAL        Normal     Normal        True             True        -0.0388
 31826026338290      1500JS       normal_clear       NORMAL         NORMAL        Normal     Normal        True             True         0.2931
 12268367291442      0500UB    anomaly_extreme      ANOMALY        ANOMALY     High Risk  High Risk        True             True         2.3041
 70720687780804      1100DS    anomaly_extreme      ANOMALY        ANOMALY     High Risk Suspicious        True            False         1.6925
 58515538325233      2602XD       normal_clear       NORMAL         NORMAL        Normal     Normal        True             True         1.0562
 58503334356532      4900VR    anomaly_extreme      ANOMALY        ANOMALY     High Risk  High Risk        True             True         2.4596
 30664536219307      4500NJ    anomaly_extreme      ANOMALY        ANOMALY     High Risk  High Risk        True             True         2.3726
 95310769629104      0505YJ       normal_clear       NORMAL         NORMAL        Normal Suspicious        True            False         1.5055
 14190538133204      4500QC       normal_clear       NORMAL         NORMAL        Normal Suspicious        True            False         1.3791
 36964394549394      1100DS  normal_borderline       NORMAL         NORMAL    Suspicious     Normal        True            False         0.0616
```

4. CONCLUSIONS

The Healthcare Fraud Anomaly Detection System successfully shifts medical fraud prevention from a reactive "Pay and Chase" model to a proactive, real-time defense. By abandoning easily bypassed static rules and implementing a hybrid machine learning architecture, the project delivers a fast, accurate, and highly scalable auditing tool.

Technical Architecture

The core achievement is the successful fusion of spatial and temporal AI. By utilizing an Isolation Forest to evaluate static document details, while simultaneously deploying an LSTM Autoencoder to monitor the doctor's historical timeline, the system catches complex behaviors that single-model systems miss. Its decoupled microservice architecture ensures heavy deep learning computations run flawlessly on standard computers without freezing the dashboard.

Operational Performance

In healthcare auditing, false alarms destroy user trust. Rigorous stress testing proved the system's operational viability. By tuning the decision threshold to the 92nd percentile, the system achieved 93.00% accuracy and a remarkable 98.44% precision score. This elite precision guarantees that when a claim is flagged as "High Risk," it is almost certainly fraudulent, allowing hospitals to confidently auto-clear "Normal" claims and reduce wasted administrative hours.

Explainability and Trust

Legally denying a medical payment requires concrete proof; an auditor cannot act simply because a computer guessed a claim was bad. By embedding SHAP directly into the user interface, the system removes the "black box" stigma of AI. It generates plain-English sentences and visual charts to show exactly which data fields triggered an alarm. Ultimately, this project proves that complex AI can be wrapped in a transparent, user-friendly dashboard for immediate enterprise deployment.

5. FURTHER DEVELOPMENT OR RESEARCH

While the current system architecture successfully proves the viability of hybrid anomaly detection, transitioning this prototype into a national-scale production environment requires specific operational upgrades. Future development should focus on four primary pillars:

- **Algorithmic Recall Enhancement:** The current model is conservatively tuned to prioritize precision, successfully eliminating false alarms. Future research must focus on improving the algorithm's "recall" metric. By experimenting with more advanced deep learning architectures— such as Transformer networks for temporal sequence modeling—the system could capture highly subtle, edge-case anomalies without sacrificing its elite precision.

- **Enterprise Database Migration:** The prototype utilizes a local SQLite database, which is ideal for demonstration but insufficient for enterprise workloads. The immediate next step is migrating the data tier to a distributed, cloud-managed relational database (such as Amazon RDS PostgreSQL). This is absolutely critical to handle the massive concurrent read/write throughput required when processing millions of nationwide healthcare transactions daily.

- **Security and Regulatory Compliance:** Healthcare data is heavily regulated by legal frameworks like HIPAA. Before this system can be deployed to live hospital auditors, strict security protocols must be implemented at the API gateway. Future development must integrate OAuth2 user authentication, Role-Based Access Control (RBAC), and immutable audit logs that securely track exactly which human user reviewed or overrode an AI decision.

- **Continuous Active Learning:** Currently, the operational risk zone thresholds (Normal, Suspicious, High Risk) are statically calibrated based on training data. A crucial enhancement is the implementation of a "Human-in-the-Loop" active learning pipeline. As human auditors investigate claims using the dashboard, the system should capture their final verdicts and use that feedback to dynamically retrain itself, constantly adapting to newly invented fraud strategies.

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7. APPENDIX

This section contains technical artifacts detailing the raw JSON API communication contract and the backend Swagger UI framework.

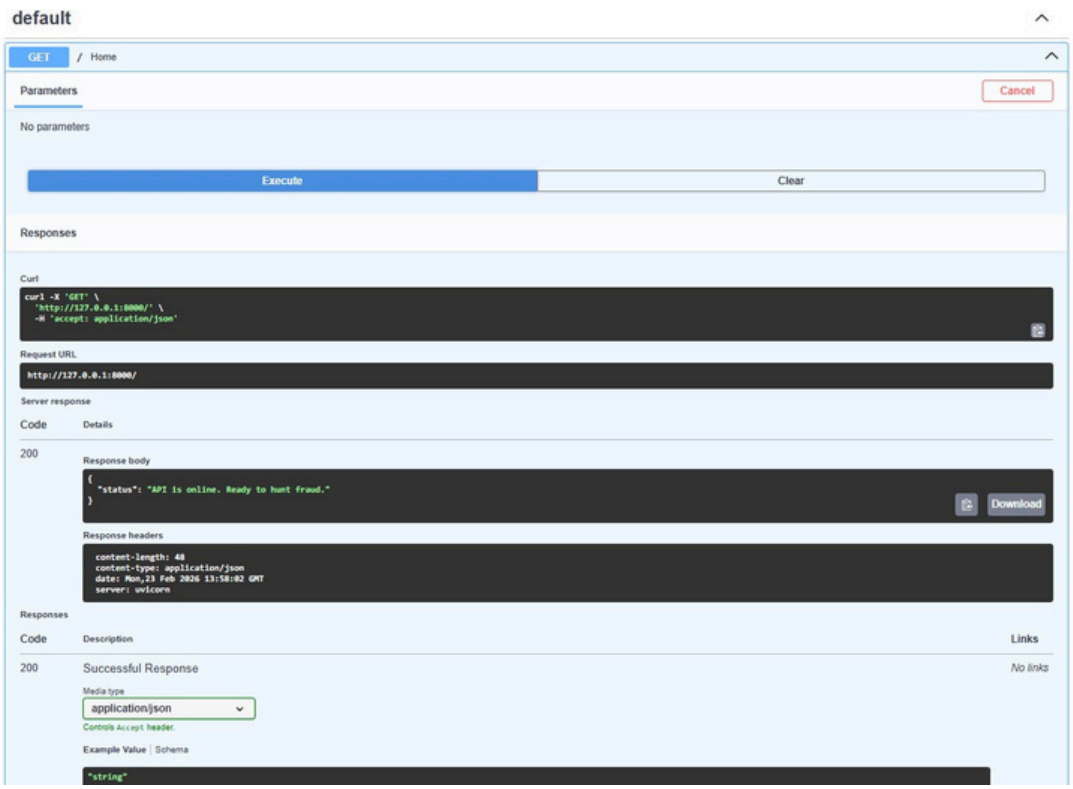
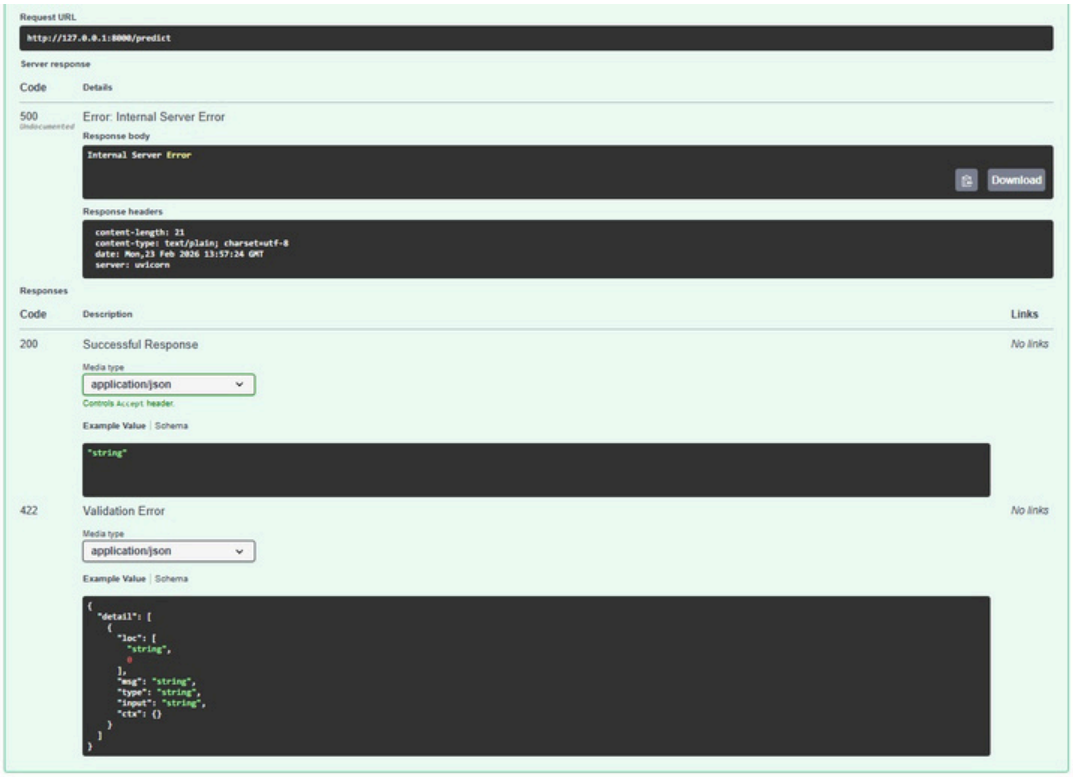
A. Sample API Input JSON Payload


```
{
  "transaction_id": 89509112315050,
  "provider_id": "2300YM",
  "claim_type": "Outpatient",
  "billing_amount": 947.25,
  "deductible_context": 0.0,
  "coinsurance_context": 40.0,
  "service_start": "2008-12-27 09:00:00",
  "service_end": "2008-12-27 10:00:00",
  "patient_age": 45,
  "diagnosis_context": "V222",
  "has_diabetes": 1,
  "has_chf": 0,
  "has_cancer": 0,
  "has_copd": 1,
  "primary_procedure": "0",
  "treatment_group": "0",
  "patient_id": "B55D152A96663E01",
  "admission_date": "2008-12-27",
  "gender_context": 1,
  "demographic_context": 1,
  "primary_service": "0"

}
```

B. API Gateway & Error Handling (Swagger UI)

API Health Check (200 OK)



API Error Handling Architecture (422 / 500)